Pagination

- Pagination is the process of dividing a large body of content into smaller, manageable pages or sections for easier viewing and navigation, both in digital and print media.
- In digital contexts like websites, it typically involves displaying a limited number of items per page, with controls like "Next," "Previous," and page numbers allowing users to move between sets of content.
- This user interface pattern makes large datasets and long documents less overwhelming and improves loading times and user experience.

Spring boot pagination

- In Spring Boot, the most common and robust way to implement pagination is by using Spring Data JPA's **Pageable and Page interfaces**.
- These provide an efficient and easy-to-use abstraction for handling large datasets by fetching data in smaller, manageable chunks.
- Pageable

  An interface used to define pagination information, including the page number (zero-indexed) and the number of records per page (page size). It can also encapsulate sorting criteria.
- Page

  An interface representing a single page of data returned from a paginated query. It contains the list of elements for the current page, along with metadata such as the total number of pages, total elements, and whether it's the first or last page.

PageRequest and Sort

- PageRequest
  - A concrete implementation of Pageable that is commonly used to create Pageable instances.
  - create a PageRequest using PageRequest.of(pageNumber, pageSize, sortCriteria).
- Sort
  - An interface used to define sorting logic, specifying fields to sort by and the sort direction (ascending or descending).
  - Sort.by(sortDirection, propertyName) allows specifying the sorting order (ascending or descending) and the field(s) to sort by.
- PagingAndSortingRepository (or JpaRepository):
  - These interfaces, extended by your custom repositories, provide methods that accept Pageable as a parameter, allowing to fetch paginated and sorted data directly from the database.

Example

```
@Repository
public interface ProductRepository extends JpaRepository<Product, Long> {
}
class Product {
   @Id
   @GeneratedValue(strategy = GenerationType.IDENTITY)
   private Long id;
   private String name;
   private double price;
}
```

Service file

```
@Service
public class ProductService {

   @Autowired
   private ProductRepository productRepository;

   public Page<Product> getAllProduct(Pageable pageable){
      return  productRepository.findAll(pageable);
   }

   public Page<Product> getPaginatedProduct(int page, int size, Sort sortBy) {
      Pageable pageable = PageRequest.of(page, size, sortBy);
      return productRepository.findAll(pageable);
   }

}
```

main

```
@SpringBootApplication
public class DemoApplication implements CommandLineRunner {
    public static void main(String[] args) {
        SpringApplication.run(DemoApplication.class, args);
    }
    @Autowired
    ProductRepository productRepository;
    @Autowired
    private ProductService productService;
    @Override
    public void run(String... args) throws Exception {
        var list1 = Arrays.asList(
                new Product[]{
Product.builder().name("biscuit").price(40).build(),
Product.builder().name("biscuit").price(40).build(),
Product.builder().name("biscuit").price(40).build(),
Product.builder().name("biscuit").price(40).build(),
productRepository.saveAll(list1)
productRepository.saveAll(list1)
productRepository.saveAll(list1)

 Pageable pageable = PageRequest.of(1, 10);
        System.out.println("using pagination ");
Page<Product> myEntitiesPage = productService.getAllProduct(pageable);

myEntitiesPage.getContent().forEach(System.out::println);
        Page<Product> page = productService.getPaginatedProduct(0, 2, Sort.by("name").ascending());

page.getContent().forEach(System.out::println);
```

controller

```
@RestController
@RequestMapping("/products")
public class ProductController {
   @Autowired
   private ProductService productService;

   @GetMapping
   public Page<Product>
getProduct(Pageable page){
      return
productService.getAllProduct(page);
   }
}
```

http://localhost:8080/products?page=0&size=10

http://localhost:8080/products?page=1&size=10

http://localhost:8080/products?page=2&size=10

http://localhost:8080/products?page=0&size=11&sort=id
,desc